Assignment 4

Wind Farm Cost Analysis Using FLORIS + LandBOSSE

Chinmay S Patil

1. Question 1: Wind Rose

The wind rose was constructed from the provided `sample_time_series.csv` time series (244 080 records) using the FLORIS `TimeSeries` and `to_WindRose` workflow. For the initial visualisation the wind direction was binned in 30° steps and the wind speed in 2 m/s steps.

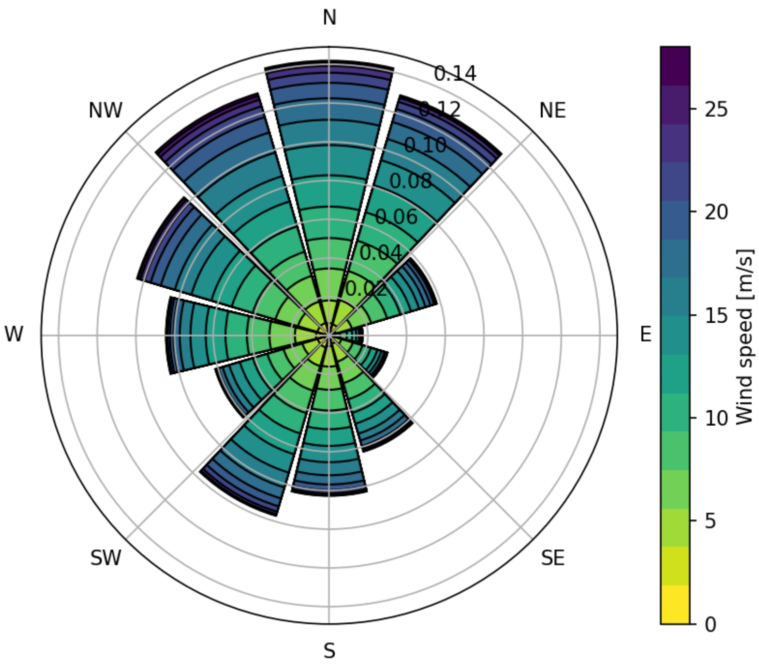

Figure 1: Wind rose of the site (direction bin 30°, speed bin 2 m/s).

2. Question 2: AEP and Wind Farm Efficiency

2.1 Setup

For the AEP computation the wind rose was re-discretised to finer bins of 1 m/s (speed) and 1° (direction), following the hint given in the FLORIS example files `003_wind_data_objects.py` and `006_get_farm_aep.py`.

The three IEA 3.4 MW turbines were arranged in a straight north-to-south line at coordinates $(x_i,\ y_i) = (0,\ i \times 5D)$ for $i = 0, 1, 2$, where $D = 130\,m$ is the rotor diameter, giving 5D spacing.

The Gauss Curl Hybrid (GCH) wake model with default parameters was used together with a wind-shear exponent of $\alpha = 0.14$ and turbulence intensity $\mathrm{TI} = 6\,\%$.

2.2 Results

```
AEP (with wake):      75.603 GWh/yr
AEP (no wake):        76.909 GWh/yr   (approx.)

Capacity Factor:      71.1 %
Wind Farm Efficiency (P / P_no_wake):  98.3 %
```

3. Question 3: LandBOSSE – Balance-of-Plant and LCOE

3.1 LandBOSSE configuration

The `iea36_120_project_data` template was selected and the following parameters were updated to match the assignment specification:

Parameter	**Value**	**Unit**
Turbine rating	3.37	MW
Hub height	120	m
Rotor diameter	130	m
Number of turbines	3	–
Turbine spacing	5	$\times D$
Fuel cost	5.19	USD/gal
Line frequency	50	Hz
Array cable voltage	30	kV
Construction time	12	months
Rated thrust (computed)	625.8	kN

The rated thrust was computed from the FLORIS thrust coefficient at the rated wind speed:

$$C_{T,\text{rated}} = 0.8015, \qquad T = \tfrac{1}{2}\,\rho\,A\,v_{\text{rated}}^2\,C_T = 625.8\,kN.$$

3.2 Balance-of-Plant costs

```
BoP Cost (total):      $6,820,508

ManagementCost       $1,519,250
GridConnectionCost   $1,429,498
FoundationCost       $1,095,121
ErectionCost         $1,008,541
SitePreparationCost    $897,342
TransportCost          $624,000
DevelopmentCost        $150,000
CollectionCost          $96,755
SubstationCost               $0
```

3.3 Turbine capital cost and ICC

```
Turbine cost (@ $1.3 M/MW): $13,143,000
ICC (Turbine + BoP):          $19,963,508
```

3.4 LCOE calculation

The LCOE was evaluated with the annuity method:

$$\mathrm{LCOE} = \frac{\mathrm{ICC}}{\mathrm{AEP} \cdot A} + \frac{C_{\mathrm{O\&M}} + C_{\mathrm{land}}}{\mathrm{AEP}},$$

where the annuity factor for $r = 4\,\%$ and $n = 20\,\mathrm{yr}$ is

$$A = \frac{(1+r)^n - 1}{r\,(1+r)^n} = 13.590.$$

```
Annuity factor (4%, 20 yr):  13.590

Annual O&M cost:   $1,209,642   (0.016 $/kWh * AEP)
Annual land rental:  $202,200   (20,000 $/MW-yr * 10.11 MW)

LCOE:  0.0381 $/kWh  =  38.10 $/MWh
```

4. Question 4: LCOE Literature Comparison and Cost Breakdown

4.1 Literature comparison

The computed LCOE of 38.10 $/MWh falls within the typical onshore wind range of 33–70 $/MWh reported by IRENA and IEA for global projects. The value is towards the lower end of the European range (40–60 $/MWh) partly because the site has a high mean wind speed, yielding a large AEP relative to the installed capacity.

4.2 Annualised cost breakdown

```
Turbine Investment      $   967,085 /yr   33.6 %
Balance-of-Plant        $   501,865 /yr   17.4 %
O&M                     $ 1,209,642 /yr   42.0 %
Land Rental             $   202,200 /yr    7.0 %
Total                   $ 2,880,792 /yr  100.0 %
```

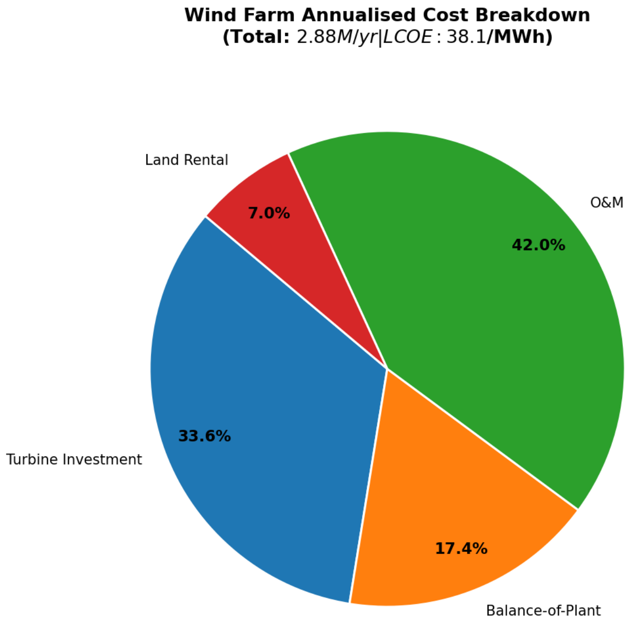

Figure 2: Annualised cost breakdown of the 3-turbine wind farm (total: $2.88 M/yr, LCOE: 38.1 $/MWh).

O&M dominates at 42 %, which is high for a three-turbine project because the fixed overhead (management, grid connection) is spread over a small installed capacity; larger farms benefit from economies of scale.

5. Question 5: Effect of Increasing Spacing from 5D to 8D

The turbine layout was updated to 8D spacing ($(x_i, y_i) = (0,\ i \times 8D)$) and both FLORIS and LandBOSSE were re-run. The substation position was adjusted accordingly (500 m perpendicular to the centre of the turbine row).

```
5D          8D          Relative difference w.r.t. 5D
AEP:    75.603 GWh/yr  76.301 GWh/yr     Delta = +0.92 %
LCOE:   38.10  $/MWh   37.94  $/MWh      Delta = -0.42 %
```

Increasing the spacing from 5D to 8D reduces wake losses (AEP rises by 0.92 %), while the BoP cost increase due to longer cable runs is small, leading to a marginal LCOE reduction of 0.42 %. The gain is modest because the three-turbine layout already operates mostly out of full wake alignment, and 5D provides sufficient recovery for the dominant wind directions at this site.

6. Question 6: Effect of a Reduced Discount Rate

If the discount rate decreases by 10 % (from $r = 4\,\%$ to $r = 3.6\,\%$), the annuity factor increases, meaning future revenues are discounted less steeply and the amortised capital cost is spread more favourably.

```
LCOE @ 4.0 %:   38.104 $/MWh
LCOE @ 3.6 %:   37.422 $/MWh
Delta:           -1.790 %
```

A 10 % relative reduction in the discount rate translates into a 1.79 % reduction in LCOE. Capital-intensive projects are more sensitive to the financing rate than fuel-cost-driven plants; however, for a modest three-turbine farm dominated by O&M costs the sensitivity is still limited.

7. Question 7: AEP Improvement Strategy

7.1 Proposed method: staggered layout

The turbines were repositioned from a pure north-to-south line to a staggered arrangement in the cross-wind direction:

$$(x_i, y_i) = \big(0,\ +5D,\ -5D\big) \times \big(0,\ 5D,\ 10D\big).$$

By offsetting turbines laterally, the downstream rotors no longer sit in the centre of the upstream wake, which increases their inflow velocity for the dominant wind directions.

```
Baseline AEP (in-line):      75.603 GWh
Staggered AEP:               76.304 GWh
AEP improvement:              0.928 %
```

7.2 Discussion

The 0.93 % gain is meaningful for a small farm. At larger scale, staggered or fully optimised layouts routinely deliver 2–5 % AEP improvement over a regular grid. Other strategies that can be applied to an already-installed farm (without repositioning turbines) include:

- **Wake steering (yaw control)**: intentionally misaligning upstream turbines to deflect their wakes away from downstream rotors.

8. Question 8: IRR and Profitability Index

8.1 Revenue and cash-flow

The average electricity spot price over the project lifetime is 41.8 $/MWh.

```
Annual Revenue:       $3,160,190   (41.8 $/MWh * 75,603 MWh)
Annual O&M Cost:      $1,209,642
Annual Land Rental:   $  202,200
Annual Net Cash Flow: $1,748,348
Initial Investment:   $19,963,508
```

8.2 NPV and IRR

The Net Present Value for a constant annual cash flow C over n years at discount rate r is

$$\mathrm{NPV}(r) = C\,\frac{(1+r)^n - 1}{r(1+r)^n} - I_0.$$

The Internal Rate of Return is the root of $\mathrm{NPV}(r) = 0$, found numerically by Brent's method. The graph for the same is seen in Figure 3.

```
IRR:  6.055 %
```

8.3 Profitability Index

The Profitability Index is

$$\mathrm{PI} = \frac{\mathrm{PV(cash\ flows)}}{I_0} = 1 + \frac{\mathrm{NPV}}{I_0}.$$

```
PV of cash flows (@4%):   $23,760,617
Initial Investment:       $19,963,508
NPV @ 4%:                  $3,797,109
PI:                            1.1902
```

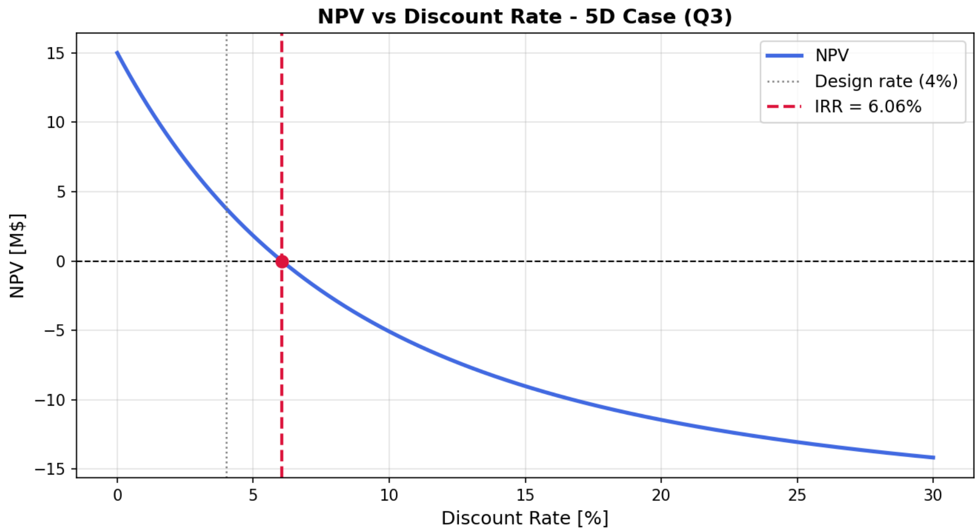

Figure 3: NPV vs. discount rate for the 5D case (Q3). The IRR of 6.06 % is marked where NPV crosses zero.

8.4 Economic assessment

The project is **economically attractive**: $\mathrm{PI} = 1.19 > 1$ and $\mathrm{IRR} = 6.06\,\% > r = 4\,\%$. The IRR exceeds the discount rate, confirming that the project creates value for investors. If, however, the electricity price were lower (e.g. at market prices closer to the LCOE), the project would break even at best. Ways to improve viability in a lower-price environment include subsidies or feed-in tariffs, a higher electricity selling price via a Power Purchase Agreement, or reducing CAPEX through economies of scale.

9. Question 9: Minimum Turbines for PI > 1

9.1 Methodology

The turbine count was varied from 1 to 10 in a straight north-to-south line with 5D spacing. For each count, FLORIS computed the AEP and LandBOSSE recomputed the BoP costs; then the PI was evaluated at $r = 4\,\%$ and $n = 20\,\mathrm{yr}$ using the same electricity price of $41.8\,\$/\mathrm{MWh}$.

9.2 Results

N	ICC [$M]	PV CFs [$M]	NPV [$M]	PI
1	11.20	6.24	−4.96	0.558
2	15.58	13.75	−1.83	0.883
3	**19.96**	**20.86**	**0.89**	**1.045**
4	24.34	27.79	3.44	1.141
5	28.73	34.62	5.90	1.205
6	33.11	41.39	8.28	1.250
7	37.49	48.13	10.64	1.284
8	41.87	54.83	12.96	1.310
9	46.25	61.51	15.26	1.330
10	50.63	68.18	17.55	1.347

The minimum number of turbines to achieve PI > 1 is **3 turbines**.

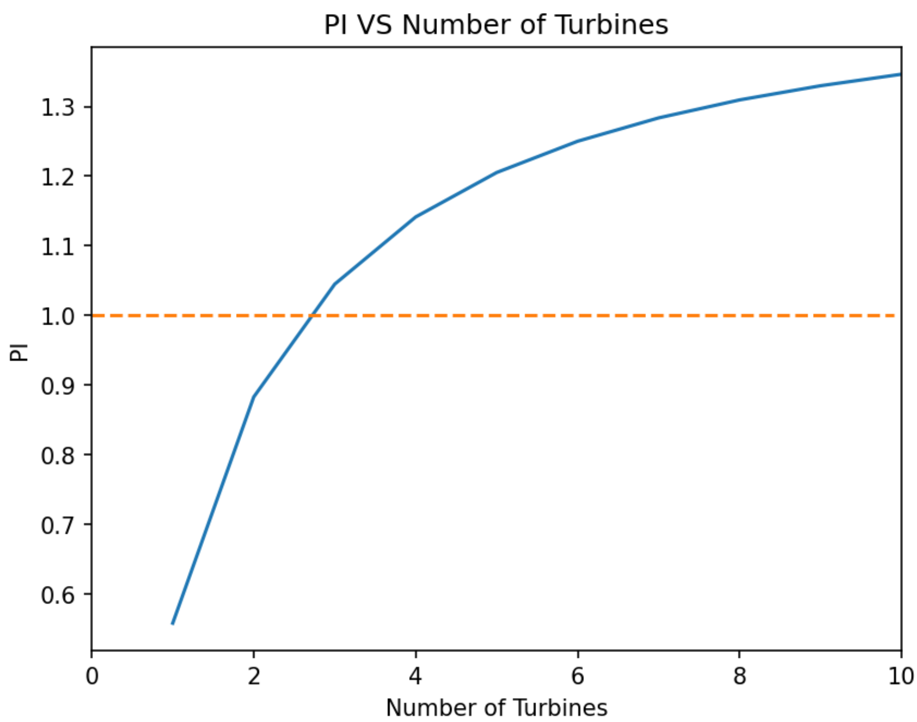

Figure 4: Profitability Index vs. number of turbines. The dashed line marks PI = 1.

9.3 Discussion

Both PI and LCOE improve monotonically as the number of turbines grows, reflecting the economies of scale in the BoP costs. The marginal benefit diminishes as N increases, so there is a practical optimum that also takes land constraints, grid capacity, and permitting into account. With only 1 or 2 turbines the project destroys value because the fixed BoP costs dominate the relatively small revenue.

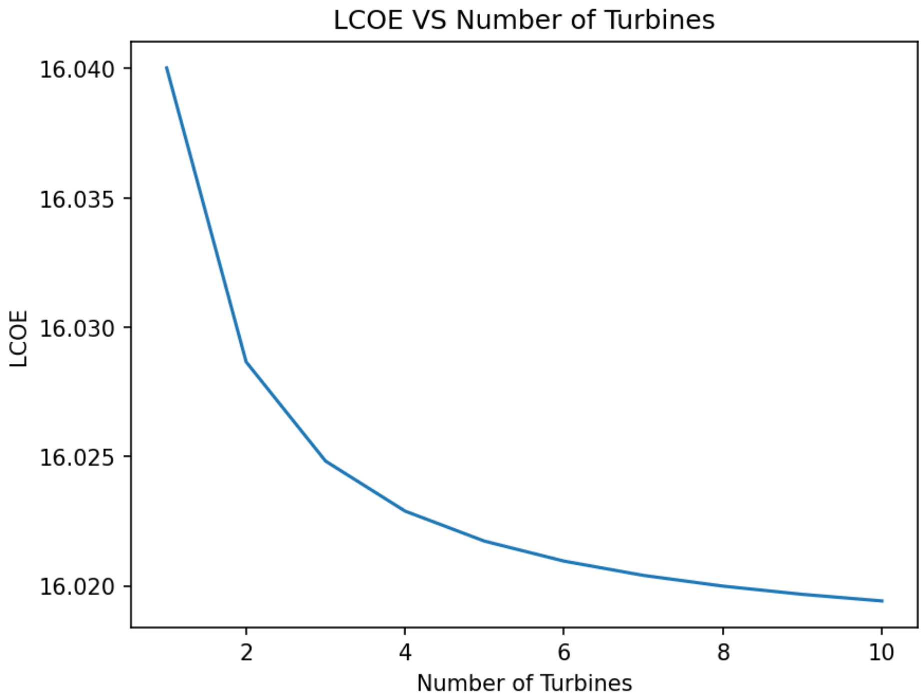

Figure 5: LCOE vs. number of turbines.